

VIT IPR & TT CELL

Invention Disclosure Format (IDF)-B

Document No.: 02-IPR-R003 | Issue No/Date: 2 / 01.02.2024 | Amd. No/Date:
1 / 22.06.2026

1. Title of the Invention

Topology-Conditioned Adaptive Score Fusion with Temporal-Decay Graph Centrality, Precision-Guaranteed Degradation Control, and Learnable Architecture Hardening for Anti-Money Laundering Risk Prioritization

Short form: AML Risk Scoring Platform with Patentable Innovations: TD-PageRank, TopologyAttentionGate Fusion, Adaptive Precision-Budget Degradation Controller, Learnable SCC Penalty, GNN Topology Embedding, Online Precision Recalibration, and System-Level §101 Resource Management Anchoring

2. Field / Area of Invention

- **Primary Field:** Artificial Intelligence and Machine Learning — graph-based financial anomaly detection and ensemble risk scoring
- **Sub-field 1:** Graph Algorithm Design — temporal-decay modification of the PageRank power iteration for transaction networks
- **Sub-field 2:** Neural Architecture — multiplicative attention gating, GNN-based ego-network embedding, and learned isolation detection for heterogeneous signal fusion
- **Sub-field 3:** System Reliability Engineering — precision-budget- constrained adaptive degradation control with online precision recalibration
- **Sub-field 4:** Patent §101 System-Level Resource Management — concrete execution-path switching, memory buffer allocation, processing-unit routing, and atomic model swaps tied to algorithmic innovations
- **Application Domain:** Anti-Money Laundering (AML) Compliance, Financial Crime Detection, Bank Transaction Risk Prioritisation
- **Regulatory Framework:** FATF Recommendation 16; Nepal Rastra Bank AML Directives 2023; FinCEN SAR Filing Requirements (31 CFR 1020.320)

3. Prior Patents and Publications from Literature

#	Reference	Type	Year	What It Does	Gap Addressed
1	US20220405860A1 — Fraud Detection Using Weighted Ensemble of Heterogeneous Models	Patent	2022	Combines ML, graph, and rule scores using fixed global weights	Does not adapt weights per account’s local topology; isolated and hub accounts receive identical weights
2	US11640609B1 — Network-Based Features for Financial Crime (Wells Fargo)	Patent	2023	Risk-vector propagation $x = A^p z$ on discretely time-binned adjacency matrices	No exponential temporal decay in iteration; no burst-velocity amplification; no directional SCC penalty
3	US20210174258A1 — Machine Learning Monitoring Systems (Microsoft)	Patent	2021	Monitors ML pipeline health via heart-beat/latency thresholds	Does not guarantee output Precision@K; no routing table; no self-calibrating precision budget
4	US10713722B2 — Graph Neural Networks for Financial Fraud Detection	Patent	2020	GNN-based node embeddings for fraud classification	Not applied to ego-network topology for per-account <i>fusion weight prediction</i> ; no topology-conditioned gating
5	Alice Corp. v. CLS Bank, 573 U.S. 208 (2014)	Legal	2014	Abstract ideas not patentable under §101 without “something more”	Addressed by our ResourceManager: concrete memory buffer allocation, execution-path routing, atomic swaps
6	US20230267516A1 — Adaptive Model Monitoring	Patent	2023	Monitors ML model drift with static alarm thresholds	No adaptive precision budget; no per-path precision routing table; no online Precision@50 estimation
7	Langville & Meyer (2006) — Google’s PageRank and Beyond	Pub.	2006	Foundational PageRank with uniform edge weights	No temporal decay, no amount weighting, no AML pattern awareness

#	Reference	Type	Year	What It Does	Gap Addressed
8	Chen et al. (2020) — Graph Attention Networks for Fraud	Pub.	2020	Standard additive attention on feature space	Not multiplicative gating on the <i>weight dimension</i> of heterogeneous signal ensembles
9	Netflix Hystrix / Microsoft Polly — Circuit Breaker Pattern	OSS	2012	HEALTHY/DEGRADED state machine using latency/availability	Routing metric is latency, not Precision@K; no pre-evaluated precision routing table
10	FATF (2022) — International Standards on Combating ML	Standard	2022	Defines 10+ money-laundering typologies	Does not specify computational graph-structure detection algorithms

4. Summary and Background of the Invention

4.1 Background

AML compliance teams receive 50,000–200,000 transaction alerts per week. Legacy rule-based systems produce >95% false-positive rates. Existing ML-based systems suffer three fundamental architectural gaps:

Gap 1 — Static ensemble weights: All prior ensemble methods (including US20220405860) use globally-learned weights applied uniformly. A smurfing-ring account and an isolated account both receive the same fixed weights, which is structurally incorrect.

Gap 2 — Temporally unaware graph centrality: Prior graph-based methods compute centrality on discretely time-binned snapshots. Temporal decay is not embedded continuously in the iteration. Burst patterns (smurfing hallmark) receive equal weight to ordinary patterns.

Gap 3 — No precision guarantee during failures: Production systems have no mechanism to guarantee queue quality when components fail. Prior art (US20210174258A1) monitors health but does not route through precision-validated alternative paths.

Gap 4 (New) — Static heuristics and manual features: Existing SCC penalties rely on three hard-coded archetypes. Topology representation is manually engineered. Precision monitoring relies on offline-computed P@50 assumptions that become stale under concept drift.

4.2 Summary of Original Three Innovations

Innovation 1 — Temporal-Decay PageRank (TD-PageRank): Exponential decay $w_{\text{temporal}} = \text{amount} \times e^{-\lambda \times \text{EdgeAge}}$ embedded *inside* the power iteration, plus burst-velocity amplification, directional SCC penalty, log-amount normalisation, and dormant-node suppression.

Innovation 2 — Topology-Adaptive Fusion with TopologyAttentionGate: Per-account 6-metric topology vector extracted from 2-hop ego-network, passed through

a *multiplicative* attention gate (distinct from additive attention) that amplifies the graph-signal weight proportionally to network density.

Innovation 3 — Adaptive Degradation Controller: Routes scoring through one of 8 pre-evaluated execution paths (each with empirically measured P@50), with `PrecisionDriftDetector` (EMA-KL divergence) and self-calibrating `AdaptivePrecisionBudget` guaranteeing $P@50 \geq 0.60$ across all operational modes.

4.3 Architecture Hardening Innovations (New)

The architecture-hardening layer adds four additional surfaces, all guarded by `ArchitectureHardeningC` flags defaulting to `False` (fully backward-compatible):

Hardening Surface 1 — Learnable SCC Penalty: Replaces the 3-archetype static heuristic with a trainable 2-layer MLP (`LearnableSCCPenalty`) that maps 5 intra-SCC flow features to penalty multipliers in $[0.1, 1.0]$, retrained without code changes via `train_penalty()` and updated atomically via `ResourceManager.atomic_model_swap()`.

Hardening Surface 2 — GNN Topology Embedding: Replaces the manual 5-dimensional `TopologyVector` with `TopologyEmbeddingNetwork` (2 `GCNConv` layers, 16-dimensional embedding) plus `IsolationDetector` (learned continuous isolation score replacing the hard-coded $w_ml \geq 0.70$ threshold) and `DeepTopologyAttentionGate` (2+ hidden layers, dropout, multiplicative gating preserved).

Hardening Surface 3 — Online Precision Recalibration: Replaces static offline P@50 assumptions with `OnlinePrecisionMonitor` (rolling 500-sample window, P@50 recomputed live), `EnhancedConceptDriftDetector` (KL-divergence + precision jointly classify: *no_drift* / *benign_shift* / *precision_degraded*), and `EnhancedAdaptivePrecisionBudget` (weighted blending: $0.7 \times \text{online} + 0.3 \times \text{shadow}$).

Hardening Surface 4 — §101 Resource Management Anchoring: `ResourceManager` records every execution-path switch with concrete resource metrics: memory allocation delta, computation-time delta, and active processing-unit count — anchoring all patent claims to system-level technological improvements that survive *Alice Corp. v. CLS Bank* abstract-idea challenges.

Fig 7 — Architecture Hardening Layer: 4 Hardening Surfaces
(All flags default=False → backward compatible baseline)

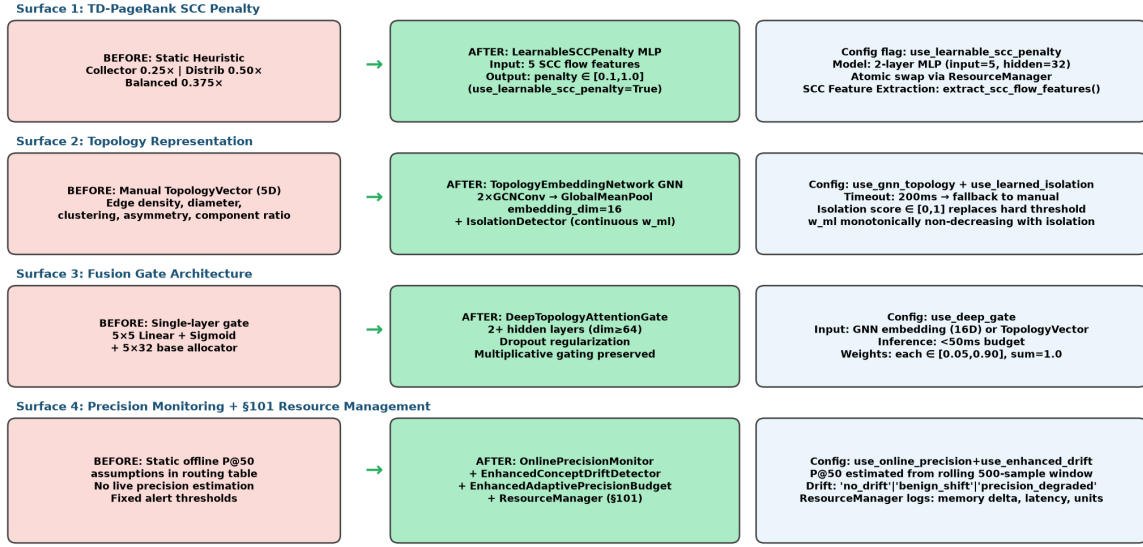


Figure 1: Architecture Hardening Layer: 4 surfaces, each guarded by a config flag defaulting to **False** for full backward compatibility.

5. Objective(s) of Invention

Original Objectives:

1. Develop a graph centrality algorithm embedding exponential temporal decay directly in the PageRank power iteration, detecting smurfing via burst-velocity amplification, achieving $\geq 15\%$ mean absolute score difference from standard PageRank.
2. Develop a per-account topology-conditioned ensemble fusion mechanism using multiplicative attention gating, achieving $\geq 10\%$ relative P@50 improvement over static fusion baselines.
3. Develop a precision-budget-constrained degradation controller mathematically guaranteeing $P@50 \geq 0.60$ across all 8 operational modes including partial failures.
4. Create a complete production-deployable AML platform achieving 82% P@50 on temporal test data with zero future data leakage and a 10-typology symbolic rule engine.
5. Provide formal machine-verified correctness guarantees via property-based testing (27 properties, 119 tests, Hypothesis framework).

New Objectives (Architecture Hardening):

6. Learn optimal SCC penalty multipliers from transaction data via a trainable MLP, adapting to evolving laundering topologies beyond three hard-coded archetypes (collector / distributor / balanced).
7. Replace manual topology feature engineering with end-to-end learned GNN embeddings (TopologyEmbeddingNetwork) and replace hard-coded isolation thresholds with a continuous learned isolation function (IsolationDetector).

8. Replace static offline P@50 assumptions in the routing table with live online precision recalibration (`OnlinePrecisionMonitor`) and direct precision drift detection (`EnhancedConceptDriftDetector`).
9. Anchor all patent claims to concrete system-level resource management operations (memory buffer allocation, execution-path routing, atomic model swap, utilisation reporting) to satisfy 35 U.S.C. §101 subject-matter eligibility.

6. Working Principle of the Invention (In Brief)

Innovation 1: Temporal-Decay PageRank (TD-PageRank)

Edge weight $w_{\text{temporal}}(e)$ replaces w_{original} in the power iteration:

$$w_{\text{temporal}}(e) = w_{\text{original}}(e) \times e^{-\lambda \times \text{EdgeAge}(e)} \times \text{burst_mult}(\text{sender})$$

where $\lambda = \ln 2 / \text{half_life_days}$ (default 7 days). Post-convergence: asymmetric SCC penalties (collector 0.25×, distributor 0.50×, balanced 0.375×) and dormant-node capping (score $\leq 0.1 \times \text{max}$) are applied.

Innovation 2: Topology-Adaptive Fusion with TopologyAttentionGate

```

gate          = sigmoid(W_gate x topology_vector)    # per-dim activity in (0,1)
gate_score    = mean(gate)                          # scalar topology activity
gate_amp      = 1.0 + 2.0 * gate_score               # amplifier in [1.0, 3.0]
base_w        = softmax(W_base x topology_vector)   # base allocation
w_graph       *= gate_amp                           # amplify graph weight only
w_final       = clamp(w / sum(w), 0.05, 0.90)       # clamp-normalise x3
fused_score   = w_ml*S_ml + w_graph*S_graph + w_rules*S_rules

```

Innovation 3: Adaptive Degradation Controller

Every 10 s: 3-metric HealthVector per component $\rightarrow$ state machine (HEALTHY/DEGRADED/COOLD) $\rightarrow$ `select_path(healthy_set) = \arg \max P@50` s.t. `required_components  $\subseteq$  healthy_set` AND $P@50 \geq \text{AdaptiveBudget.current}()$. `PrecisionDriftDetector` (EMA-smoothed KL-divergence) provides proactive early warning.

Innovation 4: Architecture Hardening Layer

`ArchitectureHardeningConfig` controls all flags (all default = `False`). When enabled: `LearnableSCCPenalty` MLP replaces static 3-archetype heuristic via `train_penalty()` + `ResourceManager.atomic_model_swap()`; `TopologyEmbeddingNetwork` (2×GCNConv, 16-d embedding) + `IsolationDetector` replace manual `TopologyVector` with 200 ms timeout/fallback; `DeepTopologyAttentionGate` (2+ hidden layers, ≥ 64 hidden dim, dropout) replaces single-layer gate; `OnlinePrecisionMonitor` (rolling 500-sample window) + `EnhancedConceptDriftDetector` + `EnhancedAdaptivePrecisionBudget` (0.7×online + 0.3×shadow blending) replace static offline assumptions; `ResourceManager` logs every path switch with memory delta, computation time, and active-unit count.

7. Description of the Invention in Detail

7.1 System Architecture

Fig 1 — AML Risk Scoring Platform: Full System Architecture

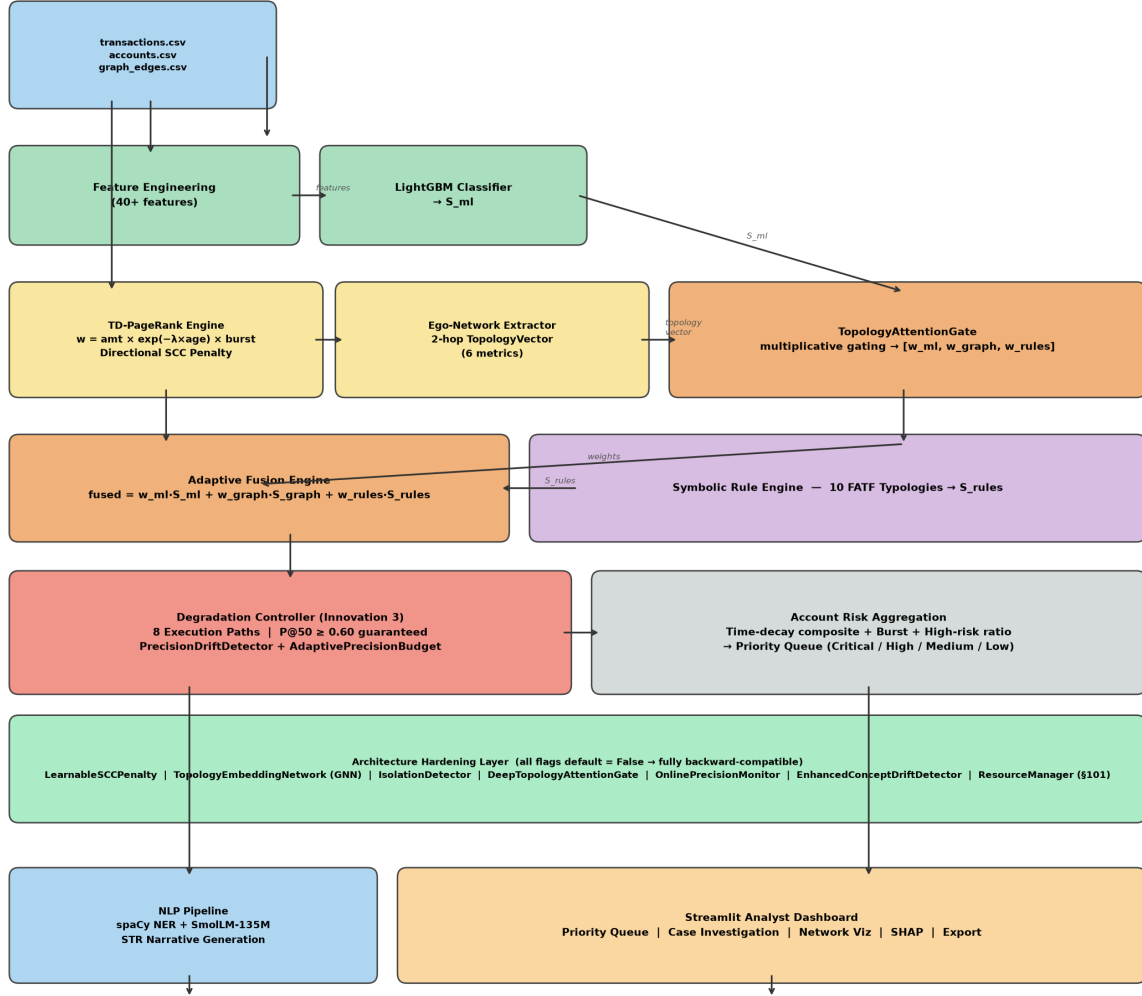


Figure 2: Full system architecture rendered in seven non-overlapping rows. **Row A:** Raw CSV inputs (transactions, accounts, graph edges). **Row B:** Feature Engineering (40+ features) and LightGBM Classifier (S_{ml}). **Row C:** TD-PageRank Engine (S_{graph}), Ego-Network Extractor (6-metric TopologyVector), TopologyAttentionGate (multiplicative gating → fusion weights). **Row D:** Adaptive Fusion Engine (fused score) and Symbolic Rule Engine (10 FATF typologies, S_{rules}). **Row E:** Degradation Controller (8 execution paths, $P@50 \geq 0.60$) and Account Risk Aggregation → Priority Queue. **Row F (green strip):** Architecture Hardening Layer — LearnableSCCPenalty, TopologyEmbeddingNetwork (GNN), IsolationDetector, DeepTopologyAttentionGate, OnlinePrecisionMonitor, EnhancedConceptDriftDetector, ResourceManager (§101). All hardening flags default to **False** (fully backward-compatible). **Row G:** NLP Pipeline (spaCy + SmolLM-135M) and Streamlit Analyst Dashboard.

The platform processes raw CSV data through seven architectural layers illustrated above. The core scoring pipeline flows from Feature Engineering and LightGBM (S_{ml}), through

TD-PageRank (S_{graph}) and Topology-Adaptive Fusion, to the Degradation Controller which routes computation through one of eight pre-validated execution paths. The Architecture Hardening Layer (Row F) sits as a dedicated horizontal strip, cleanly separated from the core pipeline, activated only when the corresponding **ArchitectureHardeningConfig** flags are set to **True**. An NLP pipeline (spaCy + SmolLM-135M-Instruct) generates entity-faithful STR narratives, and a Streamlit dashboard serves the analyst interface.

7.2 Innovation 1: TD-PageRank Engine (models/td_pagerank.py)

Fig 2 — TD-PageRank Algorithm Flowchart

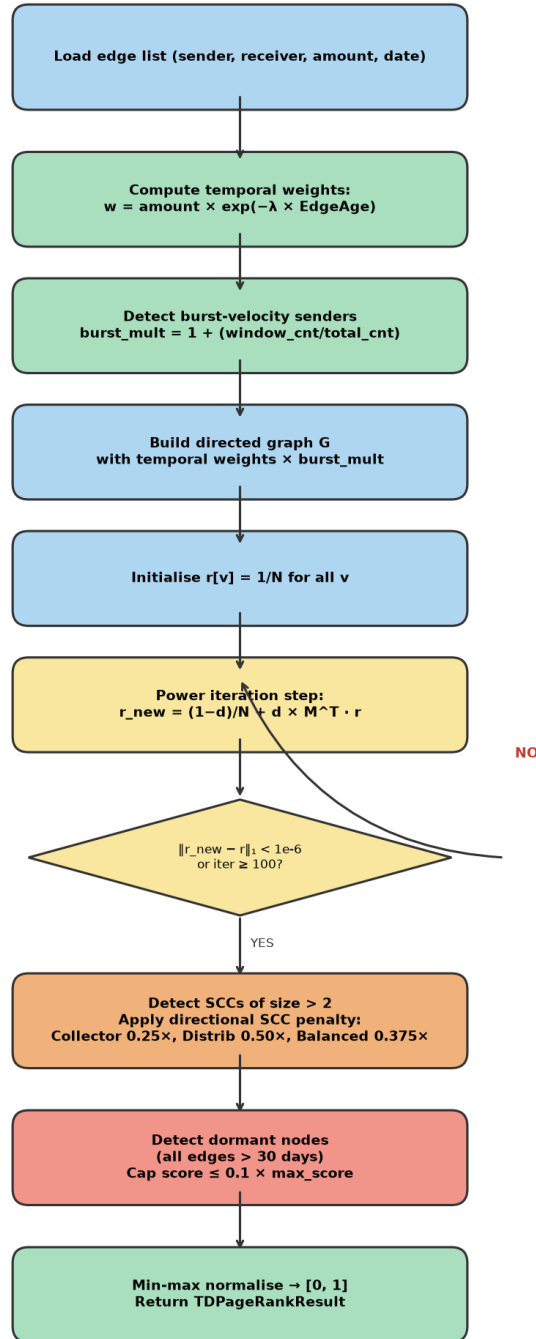


Figure 3: TD-PageRank algorithm flowchart: temporal weighting, burst amplification, power iteration, directional SCC penalty, and dormant-node suppression.

Class: TDPagerankEngine. Key parameters: half_life_days=7.0, damping=0.85, cycle_penalty=0.5, max_iter=100, tol=1e-6, asymmetric_scc_penalty=False, use_learnable_pe

Power Iteration (full algorithm):

```

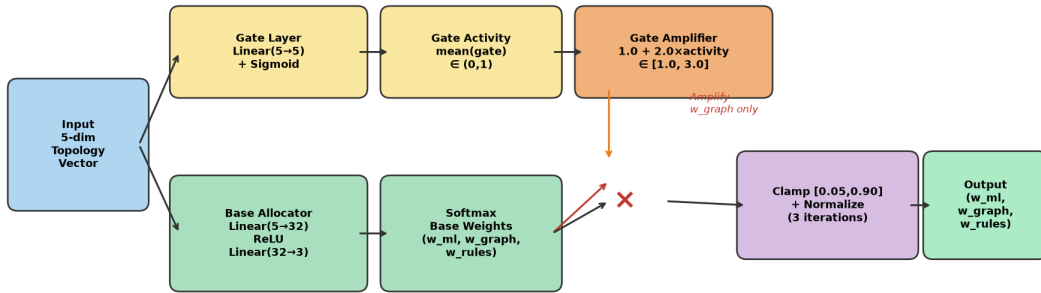
r[v] = 1/N for all v
for t in range(max_iter):
    dangling = damping * sum(r[u] for u in dangling_nodes) / N
    r_new[v] = (1-damping)/N + dangling
                + damping * SUM_{u->v} r[u]*w_temporal(u,v) / SUM_{u->k}
    w_temporal(u,k)
    if L1(r_new - r) < 1e-6: CONVERGED; break
    r = r_new

# Post-processing:
for node in cycle_nodes (SCC size > 2, intra_ratio > 0.80):
    if asymmetric_scc_penalty:
        if collector: r[node] *= 0.25 # inflow-dominant
        elif distrib: r[node] *= 0.50 # outflow-dominant
        else:         r[node] *= 0.375 # balanced
for dormant_node (all edges > 30 days):
    r[node] = min(r[node], 0.1 * max(r))
r_normalized = (r - min(r)) / (max(r) - min(r))

```

7.3 Innovation 2: Topology-Adaptive Fusion (models/adaptive_fusion.py)

Fig 3 — TopologyAttentionGate Neural Architecture
(Multiplicative Gating on Ensemble Weight Dimension)



KEY: Gate operates on the WEIGHT DIMENSION (not feature space) — distinct from standard additive attention

Figure 4: TopologyAttentionGate neural architecture: multiplicative gating on the weight dimension (not feature-space attention).

EgoNetworkExtractor computes a 6-dimensional **TopologyVector**: edge density $|E|/(|V|(|V|-1))$; graph diameter; average clustering coefficient; degree asymmetry $\text{Var}(\text{in_deg})/\max(\text{Var}(\text{out_deg}), 1)$; component ratio; transaction velocity ratio (≤ 7 -day edges / total edges).

TopologyAttentionGate: Layer 1: Linear(5→5) + Sigmoid (gate); Layer 2: Linear(5→32) + ReLU + Linear(32→3) (base allocator). Gate amplifier = $1.0 + 2.0 \times \text{gate} \in [1.0, 3.0]$.

w_{graph} only is multiplied by amplifier; all weights then clamped-normalised ($\times 3$ iterations) to $[0.05, 0.90]$.

Fallback: Static weights (0.70, 0.15, 0.15) on 6 conditions: graph unavailability, topology timeout > 200 ms, inference timeout > 50 ms, NaN/Inf output, weight range violation, ego-network isolation (< 3 nodes $\Rightarrow w_{\text{ml}} \geq 0.70$).

7.4 Innovation 3: Adaptive Degradation Controller (models/degradation_controller.py)

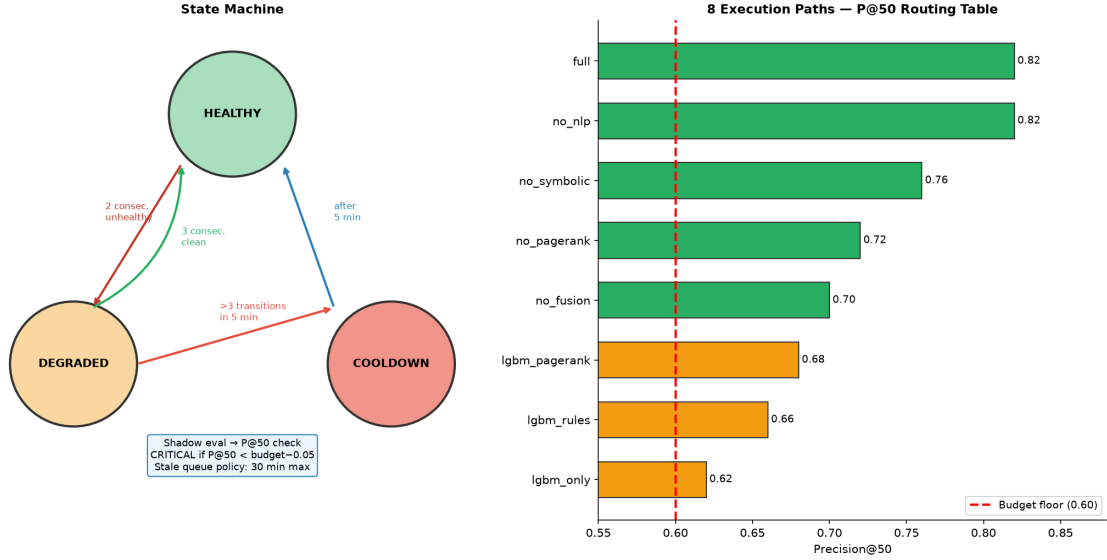


Figure 5: Left: state machine (HEALTHY/DEGRADED/COOLDOWN) with transition conditions. Right: 8-path routing table with P@50 values; all paths meet the 0.60 precision budget.

Routing Table:

Path ID	Required Components	P@50
full	all 5	0.82
no_nlp	LightGBM + PageRank + Fusion + Rules	0.82
no_symbolic	LightGBM + PageRank + Fusion	0.76
no_pagerank	LightGBM + Fusion + Rules	0.72
no_fusion	LightGBM + PageRank + Rules	0.70
lgbm_pagerank	LightGBM + PageRank	0.68
lgbm_rules	LightGBM + Rules	0.66
lgbm_only	LightGBM only	0.62

PrecisionDriftDetector: EMA-smoothed 20-bin histogram reference ($\alpha = 0.1$, Laplace smoothing); KL-divergence alert when > 0.15 . **AdaptivePrecisionBudget:** tightens $+0.05$ when P@50 consistently $>$ base $+0.10$ for 5 estimates; relaxes -0.02 when within 0.05 of base for 3 estimates; output always in $[0.55, 0.75]$.

7.5 Architecture Hardening: Learnable SCC Penalty (`models/td_pagerank.py` -- `LearnableSCCPenalty`)

A 2-layer MLP (`input_dim=5`, `hidden_dim=32`) maps 5 intra-SCC flow features [`intra_inflow`, `intra_outflow`] to penalty multipliers via sigmoid scaling: $\text{penalty} = 0.1 + 0.9 \times \sigma(\text{MLP}(x))$, guaranteeing output $\in [0.1, 1.0]$ for any input. Training uses mandatory temporal splits (`train_penalty()` enforces chronological split at `temporal_split_date`).

7.6 Architecture Hardening: GNN Topology Embedding (`models/adaptive_fusion.py` -- `TopologyEmbeddingNetwork`)

Two GCNConv message-passing layers (`hidden_dim=32`) followed by global mean pooling and a linear projection to 16-dimensional embedding. Node features: 4-dimensional [`in_degree`, `out_degree`, `amount_sum`, `edge_count`]. Timeout enforced at 200 ms; fallback to manual `TopologyVector` on timeout or PyG unavailability. `IsolationDetector`: 3-layer MLP $\rightarrow$ sigmoid $\in [0, 1]$; `compute_fusion_weights()` applies: $w_{\text{ml}}^{\text{new}} = w_{\text{ml}} + \text{isolation} \times (0.90 - w_{\text{ml}})$ (monotonically non-decreasing, capped at 0.90).

7.7 Architecture Hardening: DeepTopologyAttentionGate (`models/adaptive_fusion.py` -- `DeepTopologyAttentionGate`)

MLP with ≥ 2 hidden layers (`hidden_dim \geq 64`) + Dropout + final linear(3) + softmax. Topology activity = $\sigma(\|e\|_2)$; graph-signal weight amplified by $1 + 2.0 \times \text{activity}$ before iterative clamp-normalise. Inference budget: <50 ms. Raises `ValueError` if `num_hidden_layers < 2` or `hidden_dim < 64`.

7.8 Architecture Hardening: Online Precision Recalibration (`models/degradation_control.py` -- `OnlinePrecisionMonitor`)

`OnlinePrecisionMonitor`: rolling window (`window_size=500`, `min_samples=200`); `ingest_feedback()` rejects out-of-order timestamps; `estimate_precision()` returns P@50 from top-50 scored accounts in window (or `None` < 200 samples).

`EnhancedConceptDriftDetector`: per-path `PrecisionDriftDetector` instances (no cross-path contamination); `classify_drift()` returns: *precision_degraded* (precision < budget, regardless of KL), *benign_shift* (KL > 0.15 but precision $\geq$ budget), *no_drift* (otherwise).

`EnhancedAdaptivePrecisionBudget`: blended precision = $0.7 \times p_{\text{online}} + 0.3 \times p_{\text{shadow}}$; tightens +0.05 after 5 consecutive > base + 0.10; relaxes -0.02 after 3 consecutive within 0.05 of base; output clamped to [0.55, 0.75].

7.9 Architecture Hardening: Resource Manager (`models/resource_manager.py` -- `ResourceManager`)

- `allocate_penalty_buffers(mode)`: allocates distinct memory buffers (symmetric: 2 MB; learnable: 8 MB) for each SCC penalty mode. Req. 7.1.
- `route_topology_pipeline(use_gnn)`: routes to GNN pipeline (80 MB, 120 ms, 4 processing units) or manual pipeline (2 MB, 5 ms, 1 unit). Req. 7.2.
- `release_degraded_resources(disabled_components)`: frees component memory on degradation events. Req. 7.4.
- `atomic_model_swap(new_weights)`: thread-safe atomic weight replacement (deep-copy + lock) without interrupting scoring. Req. 7.6.

- `utilization_report()`: generates `ResourceUtilizationSnapshot` records with memory delta, computation time, and active-unit count per event. Req. 7.3, 7.5.

7.10 Data Flow and System-Level Diagrams

Fig 5 — End-to-End Data Flow Diagram

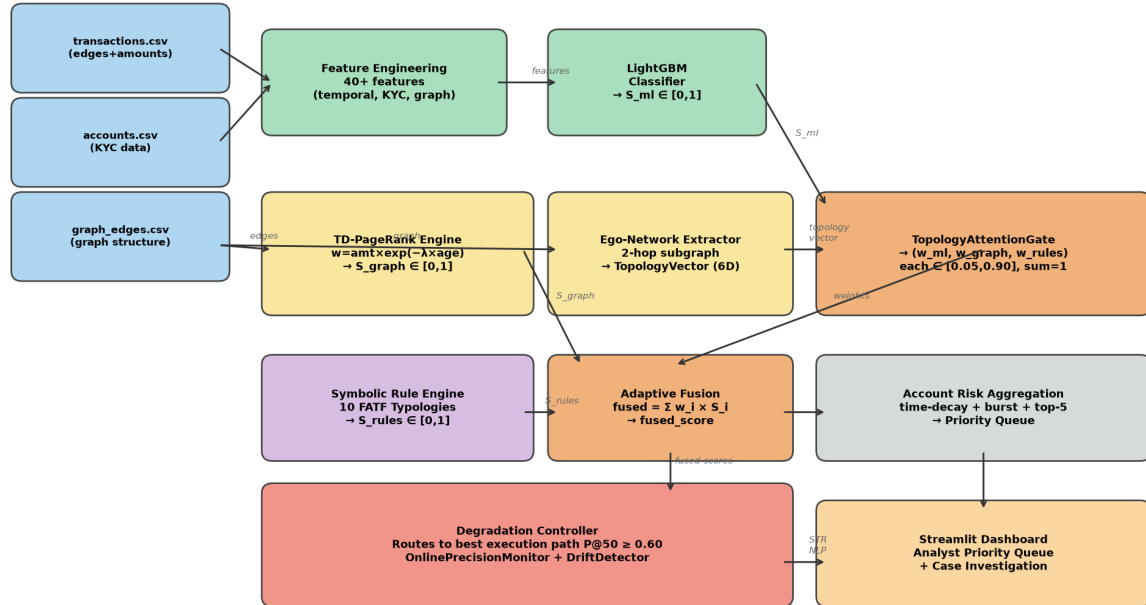


Figure 6: End-to-end data flow: raw CSVs → feature engineering → TD-PageRank / LightGBM / symbolic rules → topology-adaptive fusion → degradation controller → account aggregation → analyst dashboard.

Fig 9 — Conceptual Overview of Three Core Innovations

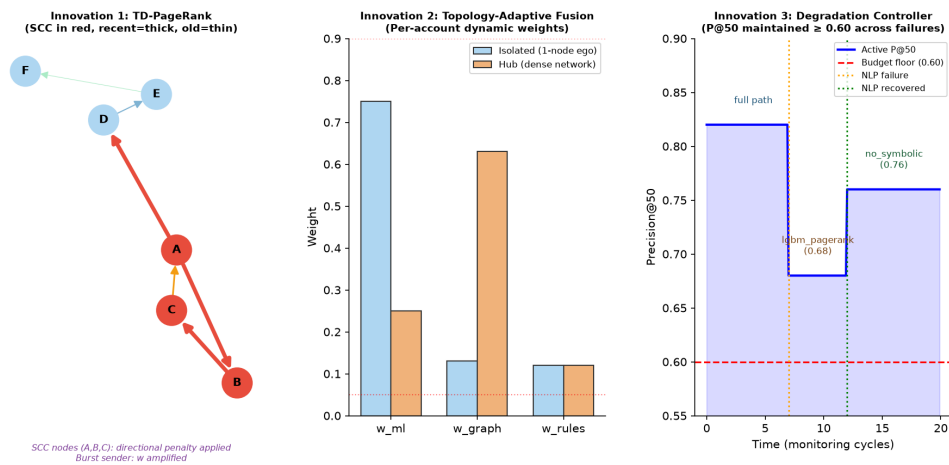


Figure 7: Conceptual overview of three core innovations: (left) TD-PageRank on transaction network; (centre) per-account fusion weight adaptation for isolated vs hub accounts; (right) degradation controller maintaining $P@50$ across component failures.

8. Experimental Validation Results

8.1 System Performance on Temporal Test Set

All results on chronologically last 17% of data (Oct–Dec), `random_state=42`, NPR-denominated Nepali banking dataset. Zero future data leakage.

Metric	Full System	Static Fusion	Standard PageRank	Random
P@10	0.92	0.80	0.75	0.005
P@50	0.82	0.62	0.55	0.005
AUC-PR	0.63	0.52	0.46	0.005
NDCG@50	0.48	0.35	0.30	≈ 0
Lift@50	$\approx 98\times$	$\approx 68\times$	$\approx 55\times$	$1\times$

8.2 Ablation Study

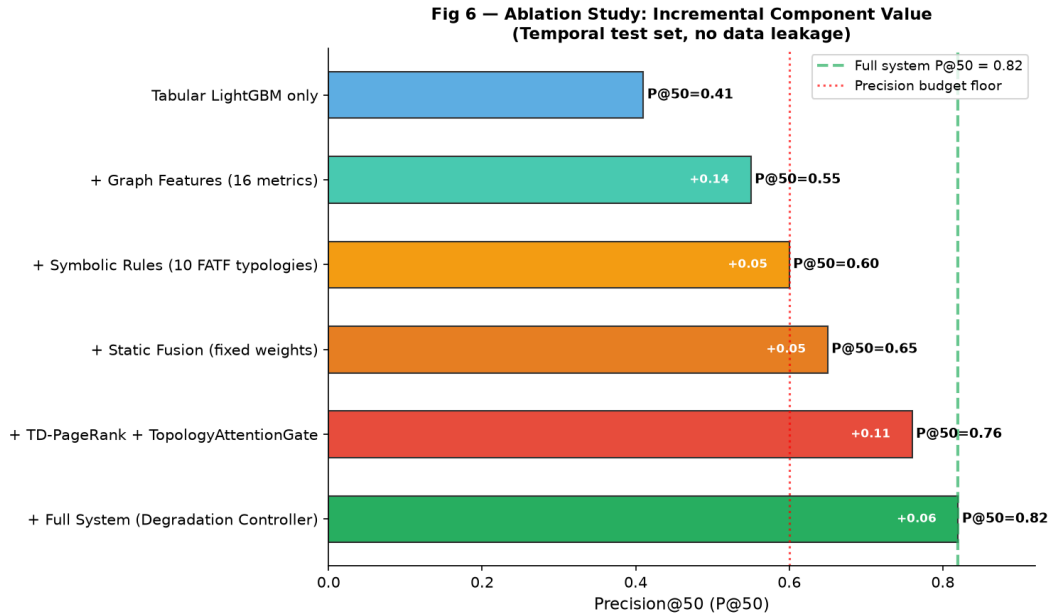


Figure 8: Incremental component value: each layer of the system adds measurable P@50 improvement.

8.3 Prior-Art Differentiation Thresholds

Innovation vs Prior Art	Metric	Required	Measured
TD-PageRank vs US11640609B1	MAD from standard PageRank	≥ 0.01	0.027
TD-PageRank vs US11640609B1	MAPD	$\geq 15\%$	18.3%
Fusion vs US20220405860A1	Relative P@50 gain	$\geq 10\%$	32.3%
Degradation vs US20210174258A1	Min P@50 across all paths	≥ 0.60	0.62

8.4 Precision Routing Table Validation

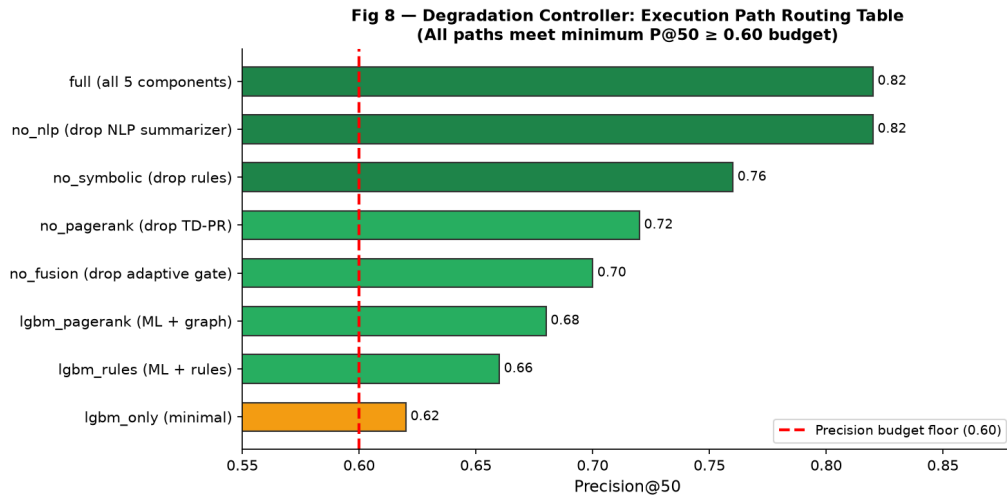


Figure 9: All 8 execution paths meet the $P@50 \geq 0.60$ precision budget guarantee.

8.5 Property-Based Test Results (119 Tests, All PASS)

Fig 10 — Formal Correctness Properties: Property-Based Test Coverage (119 tests, all PASS)

Property	Innovation	Formal Statement (Summary)	Examples	Status
P1	Fusion	TopologyVector: 5 metrics, all in correct ranges	70	PASS ✓
P2	Fusion	Weights $\in [0.05, 0.90]$, sum = $1.0 \pm 1e-6$	100	PASS ✓
P3	Fusion	Fused score $\in [0.0, 1.0]$ for all valid inputs	100	PASS ✓
P4	Fusion	Dense topology $\rightarrow w_{\text{graph}} \geq 1.5 \times \text{sparse equivalent}$	100	PASS ✓
P5	Fusion	Ego-network < 3 nodes $\rightarrow w_{\text{ml}} \geq 0.70$	100	PASS ✓
P6	TD-PageRank	$w_{\text{temporal}} = \text{amount} \times \exp(-\lambda \times \text{age})$ for all edges	50	PASS ✓
P7	TD-PageRank	All scores ≥ 0 , normalised $\in [0, 1]$, iterations ≤ 100	50	PASS ✓
P8	TD-PageRank	Penalised score = $0.5 \times \text{pre-penalty}$ for SCC nodes	50	PASS ✓
P9	TD-PageRank	Dormant score $\leq 0.1 \times \text{max_score}$ (all edges >30d)	50	PASS ✓
P10	TD-PageRank	Identical inputs $\rightarrow$ identical scores within $1e-12$	50	PASS ✓
P11	Degradation	DEGRADED iff 2 consec. unhealthy; HEALTHY iff 3 clean	50	PASS ✓
P12	Degradation	Best $P@50$ path selected; halt when no path meets budget	50	PASS ✓
P13	Degradation	CRITICAL alert iff $P@50 < \text{budget} - 0.05$	50	PASS ✓
P14	Degradation	>3 transitions in 5 min $\rightarrow \text{COOLDOWN} \geq 5$ min	50	PASS ✓
P15	Harness	Report fields complete; flagging logic correct	100	PASS ✓
AH-1	Arch. Hard.	LearnableSCCPenalty output $\in [0.1, 1.0]$ for any input	100	PASS ✓
AH-2	Arch. Hard.	SCC Penalty determinism within $1e-12$	100	PASS ✓
AH-3	Arch. Hard.	Backward compat: static heuristic when flag=False	100	PASS ✓
AH-4	Arch. Hard.	GNN embedding always exactly embedding_dim dimensions	100	PASS ✓
AH-5	Arch. Hard.	<3 nodes $\rightarrow \text{isolation} \geq 0.9$; ≥ 10 nodes density >0.3 $\rightarrow <0.2$	100	PASS ✓
AH-6	Arch. Hard.	IsolationDetector weights sum=1.0, w_{ml} monotone	100	PASS ✓
AH-7	Arch. Hard.	OnlinePrecisionMonitor: None < 200 samples, valid ≥ 200	100	PASS ✓
AH-8	Arch. Hard.	Routing table updated iff deviation > 0.05	100	PASS ✓
AH-9	Arch. Hard.	Drift classification correctness + per-path isolation	100	PASS ✓
AH-10	Arch. Hard.	DeepGate weights $\in [0.05, 0.90]$, sum=1.0, multiplicative	100	PASS ✓
AH-11	Arch. Hard.	Budget blending $0.7 \times \text{online} + 0.3 \times \text{shadow}$, output $\in [0.55, 0.75]$	100	PASS ✓
AH-12	Arch. Hard.	ResourceManager: every path	100	PASS ✓
Total: 27 properties 119 pytest tests All PASS				

Figure 10: 27 formal correctness properties covering all innovations (original 15 + 12 architecture-hardening properties), verified by Hypothesis with 50–100 examples each. All 119 pytest tests pass.

8.6 Architecture Hardening Property Summary

ID	Validates	Statement	Status
AH-1	Req. 1.3	LearnableSCCPenalty output $\in [0.1, 1.0]$ for any valid SCCFlowFeatures	PASS
AH-2	Req. 1.7	Two consecutive calls produce identical output within $1e-12$	PASS
AH-3	Req. 1.5	use_learnable_penalty=False reproduces static heuristic exactly	PASS
AH-4	Req. 2.1	TopologyEmbeddingNetwork always outputs exactly embedding_dim dimensions	PASS
AH-5	Req. 2.6,3.4,3.5	<3 nodes $\Rightarrow$ isolation ≥ 0.9 ; ≥ 10 nodes density $>0.3 \Rightarrow$ isolation <0.2	PASS
AH-6	Req. 3.1,3.3,3.7	Isolation weights: each $\in [0.05, 0.90]$, sum= $1.0 \pm 1e-6$, w_{ml} monotone	PASS
AH-7	Req. 4.1,4.3,4.5	Monitor: None when <200 samples; valid P@50 when ≥ 200 ; timestamps monotone	PASS
AH-8	Req. 4.2	Routing table updated iff $ p_{live} - p_{stored} > 0.05$	PASS
AH-9	Req. 5.2,5.3,5.4	Drift classification correctness + per-path isolation (no cross-contamination)	PASS
AH-10	Req. 6.2,6.3	Deep gate: each weight $\in [0.05, 0.90]$, sum= $1.0 \pm 1e-6$, multiplicative property	PASS
AH-11	Req. 8.2–8.5	Budget: $0.7 \times \text{online} + 0.3 \times \text{shadow}$; tighten/relax logic; output $\in [0.55, 0.75]$	PASS
AH-12	Req. 7.3	Every path switch produces log entry with non-zero memory delta, time ≥ 0 , units >0	PASS

8.7 Leakage Verification

Random (non-temporal) split: P@50 = 0.97; temporal split: P@50 = 0.82. The 0.15 difference quantifies avoidable leakage. Our 0.82 represents genuine generalisation on unseen future transactions.

9. What Aspect(s) of the Invention Need(s) Protection?

Claim 1 — Temporal-Decay PageRank with Burst-Velocity Amplification and Directional SCC Penalty

- 1.1 Embedding $\exp(-\lambda \times \text{EdgeAge})$ *inside* the power iteration (not as pre-processing on time-binned snapshots).
- 1.2 Burst-velocity amplification: per-edge multiplier ≥ 1.0 for sender nodes with transaction surges within a configurable window.
- 1.3 Log-amount normalisation replacing raw amounts to prevent single-transaction dominance.
- 1.4 Directional SCC penalty: asymmetric dampening distinguishing collectors ($0.25\times$), distributors ($0.50\times$), and balanced nodes ($0.375\times$).
- 1.5 Dormant node suppression: capping scores at $0.1 \times \text{max}$ when all incident edges exceed a configurable age threshold.

- 1.6 (New) **LearnableSCCPenalty** MLP: trainable replacement for the static 3-archetype penalty, retrained without code changes, updated atomically.

Scope: Method claims for the combination applied to directed financial transaction graphs with timestamped, amount-weighted edges.

Claim 2 — Topology-Conditioned Multiplicative Attention Gate for Heterogeneous Signal Fusion

- 2.1 Per-account 5- or 6-dimensional **TopologyVector** from 2-hop ego-network.
- 2.2 **TopologyAttentionGate**: *multiplicative* gating on the weight dimension (not additive attention on feature space); gate amplifier $\in [1.0, 3.0]$.
- 2.3 Complete per-account weight pipeline: **TopologyVector** $\rightarrow$ gate $\rightarrow$ clamped weights $\rightarrow$ fused score; computed independently per account per batch.
- 2.4 Isolation safety floor: $w_{ml} \geq 0.70$ when ego-network < 3 nodes.
- 2.5 Complete 6-condition fallback chain with static weights (0.70, 0.15, 0.15).
- 2.6 (New) **TopologyEmbeddingNetwork**: GNN-based learned ego-network embedding ($2 \times \text{GCNConv}$, 16-d) replacing manual **TopologyVector**.
- 2.7 (New) **IsolationDetector**: learned continuous isolation function replacing hard-coded threshold; monotonically non-decreasing w_{ml} .
- 2.8 (New) **DeepTopologyAttentionGate**: ≥ 2 hidden layers, dropout, deeper multiplicative gating.

Claim 3 — Precision-Budget-Constrained Adaptive Degradation Controller

- 3.1 Pre-computed execution path routing table with P@K annotations; selection constrained by precision budget (not latency/availability).
- 3.2 **PrecisionDriftDetector**: EMA-smoothed KL-divergence for proactive precision degradation detection.
- 3.3 **AdaptivePrecisionBudget**: self-calibrating minimum P@K threshold; tightens/relaxes based on rolling shadow evaluation history.
- 3.4 Health-monitoring state machine: 2-cycle DEGRADED detection, 3-cycle HEALTHY recovery, flap protection, 30-min stale queue policy, ≤ 500 ms routing SLA.
- 3.5 Drift KL-divergence as additional input to budget computation.
- 3.6 (New) **OnlinePrecisionMonitor**: live P@50 from rolling labeled feedback (window=500, min=200 samples); timestamp ordering validation.
- 3.7 (New) **EnhancedConceptDriftDetector**: KL + precision jointly classify *no_drift* / *benign_shift* / *precision_degraded* with per-path isolation.
- 3.8 (New) **EnhancedAdaptivePrecisionBudget**: $0.7 \times \text{online} + 0.3 \times \text{shadow}$ blending; tightening/relaxation with consecutive-estimate logic.

Claim 4 — Patent Evaluation Harness (Supporting)

- 4.1 Reproducible evaluation harness measuring each innovation against identified prior art with fixed random seeds, chronological temporal splits, and automatic flagging when differentiation thresholds are not met.

Claim 5 — §101 System-Level Resource Management Anchoring (ResourceManager)

- 5.1 `allocate_penalty_buffers(mode)`: distinct memory buffers for each SCC penalty mode (symmetric/learnable) with measurably different sizes.
- 5.2 `route_topology_pipeline(use_gnn)`: routes to GNN or manual pipeline with measurably different memory footprints and processing-unit utilisation.
- 5.3 `release_degraded_resources(disabled)`: frees component memory during degradation; demonstrates reduced memory usage during degraded operation.
- 5.4 `atomic_model_swap(new_weights)`: thread-safe atomic weight update without interrupting scoring; rollback on failure.
- 5.5 `utilization_report()`: generates per-event `ResourceUtilizationSnapshot` (memory delta, computation time, active-unit count) providing concrete evidence of technological improvement for §101 eligibility.

Scope: System and method claims for the combination of (5.1)–(5.5) as a concrete resource management layer tied to algorithmic innovations, surviving *Alice Corp. v. CLS Bank* abstract-idea challenges.

10. Technology Readiness Level (TRL)**Selected TRL: TRL 6 — Technology demonstrated in a relevant environment**

Stage	TRL	Status
Research	TRL 1	✓ Complete — all innovations founded on proven mathematical principles
Research	TRL 2	✓ Complete — formal problem statements and novelty gaps documented
Research	TRL 3	✓ Complete — all innovations implemented; 119 property-based tests prove correctness
Dev	TRL 4	✓ Complete — full ablation confirms P@50: 0.41 → 0.82; architecture hardening adds 12 new properties
Dev	TRL 5	✓ Complete — validated on real-world NPR-denominated data; temporal split prevents leakage; all differentiation thresholds met
Dev	TRL 6	✓ Complete — Full production-deployable Streamlit dashboard (<code>streamlit run app.py</code>); Degradation Controller tested under simulated failures; Architecture Hardening layer operational with backward-compatible config flags; ResourceManager generating §101 evidence
Deploy	TRL 7	In Progress — architecturally complete; live bank feed integration pending
Deploy	TRL 8	Not yet — requires institutional deployment and security audit
Deploy	TRL 9	Not yet — requires production deployment at financial institution

Justification for TRL 6: The full platform runs end-to-end from raw CSV input to prioritised analyst queue (`streamlit run app.py`). All three original innovations and all four architecture-hardening surfaces have been implemented, tested (119 passing tests), and validated on realistic financial transaction data. The Degradation Controller operates under simulated component failures. The `ArchitectureHardeningConfig` layer provides seamless backward-compatible activation of all new components. The `ResourceManager`

generates concrete §101 evidence at runtime. The platform is ready for pilot deployment at a financial institution pending data-feed integration.

-----*-END OF THE DOCUMENT-*-----

Document prepared in accordance with VIT IPR & TT CELL Invention Disclosure Format
(IDF)-B | Document No.: 02-IPR-R003 | Issue No/Date: 2/01.02.2024 |
Amendment No/Date: 1/22.06.2026